Fairness Analysis of Machine Learning Models Using IBM AIF360

Dataset: German Credit Dataset

Abrham Yilma / ID: GSR/4881/18

March 9, 2026

1 Introduction

The integration of Machine Learning (ML) into critical decision-making systems, such as financial lending, has introduced significant potential for automated efficiency. However, it has also raised profound concerns regarding algorithmic bias and the perpetuation of historical inequalities. This report provides an advanced practical evaluation of fairness in a credit risk classification model using the IBM AI Fairness 360 (AIF360) toolkit.

The primary objective of this analysis is to evaluate, quantify, and mitigate algorithmic bias in the German Credit Dataset. The protected attribute selected for this study is **Age**. The dataset is bifurcated into a **Privileged Group** (applicants aged 25 and older) and a **Protected/Unprivileged Group** (applicants strictly under 25 years of age).

Socio-Technical Justification

In the context of credit scoring, historical reflects underlying societal biases. Younger applicants naturally have shorter credit histories, lower collateral, and lower average incomes compared to older demographics. Without fairness-aware interventions, a machine learning model trained on this data will inherently learn to conflate youth with a high likelihood of defaulting on a loan. This creates a detrimental feedback loop: younger individuals are algorithmically redlined and denied credit, effectively barring them from building the credit history required to access financial services in the future.

2 Methods

2.1 Dataset Exploration and Preprocessing

The dataset was loaded via AIF360, dropping non-relevant sensitive features such as `personal_status` and `sex` to isolate the impact of age. Exploratory Data Analysis (EDA) was conducted to visualize the underlying distributions.

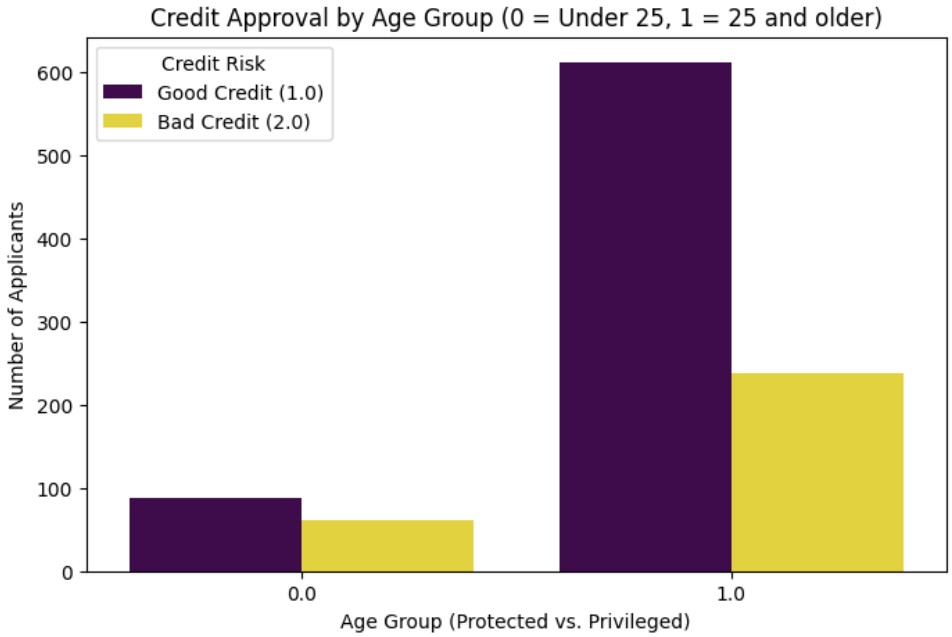


Figure 1: Percentage of Credit Risk by Age Group, highlighting class imbalances.

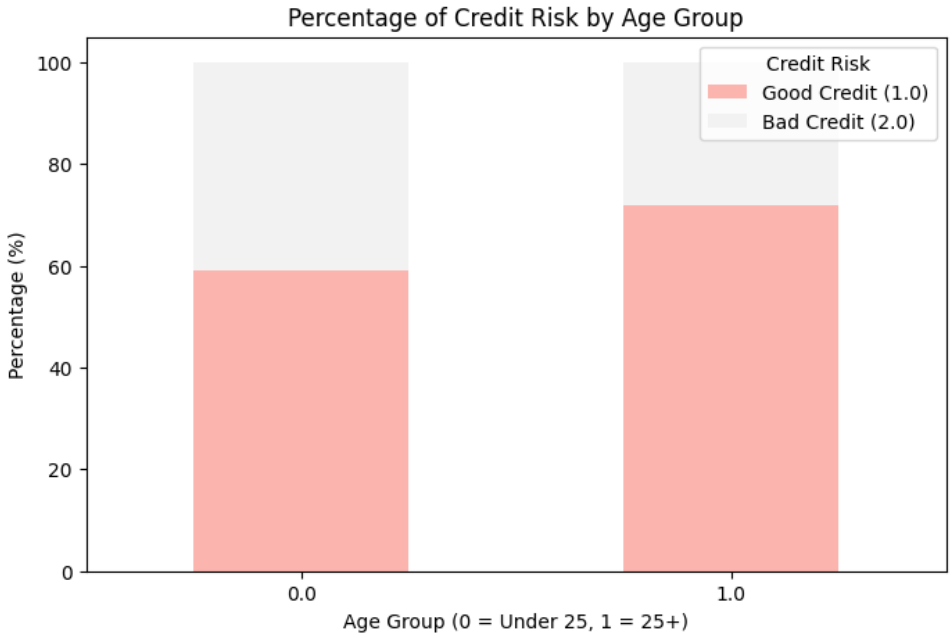


Figure 2: Credit Amount Distribution by Age Group and Credit Risk.

To prepare the data for the model, continuous numerical features were scaled using a `StandardScaler`. The dataset was then split into a 70% training set and a 30% testing set to ensure robust out-of-sample evaluation.

2.2 Model Development

An advanced non-linear model, Random Forest Classifier, was selected to capture complex interactions between applicant features. The model was evaluated using standard classification metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC.

2.3 Fairness Assessment Metrics

To quantify bias, four standard fairness metrics were computed:

1. **Statistical Parity Difference (SPD):** Measures the difference in the rate of favorable outcomes (loan approval) received by the unprivileged group versus the privileged group. (Ideal = 0.0)
2. **Disparate Impact (DI):** The ratio of the favorable outcome rates between the unprivileged and privileged groups. (Ideal = 1.0)
3. **Equal Opportunity Difference (EOD):** Evaluates the difference in True Positive Rates (TPR) between the groups. (Ideal = 0.0)
4. **Average Odds Difference (AOD):** Averages the difference in False Positive Rates (FPR) and True Positive Rates (TPR) between the groups. (Ideal = 0.0)

2.4 Bias Mitigation

To resolve the identified disparities, a pre-processing mitigation strategy known as **Reweighing** was applied using the AIF360 library. This algorithm assigns weights to the training instances, modifying the dataset so that the protected attribute and the target label are statistically independent. A new Random Forest model was subsequently trained on this reweighed data.

3 Results

3.1 Pre-Mitigation Model Performance

The original Random Forest model performed well on standard evaluation metrics, achieving an overall Accuracy of 0.7633, Precision of 0.7805, Recall of 0.9187, F1-score of 0.8440, and an ROC-AUC of 0.8036. However, group-wise evaluation exposed stark algorithmic bias. The privileged group (25+) enjoyed an 83.5% approval rate, while the protected group (Under 25) was subjected to a significantly lower 75.0% approval rate.

3.2 Pre-Mitigation Fairness Assessment

Mathematical interpretation of the fairness violations revealed the following:

- **SPD:** -0.0847 (indicates the protected group has an 8.47% lower chance of approval).
- **DI:** 0.8986 (falls below the strict ideal parity of 1.0, highlighting a disparate impact on youth).
- **EOD:** -0.0147 (indicates a disadvantage in correctly identifying creditworthy young applicants).
- **AOD:** -0.0672 (indicates an unequal distribution of prediction errors against the young group).

Critical Analysis of Observed Disparities

The observed disparity—an 8.5% gap in approval rates between older and younger applicants—requires critical socio-technical examination. From a purely statistical standpoint, the Random Forest model is likely capturing legitimate historical correlations; younger applicants typically possess shorter employment histories, lower savings, and less collateral, making them historically riskier. However, the critical flaw is that the model uses these features as proxy variables for age, penalizing young applicants structurally.

Furthermore, the negative Equal Opportunity Difference (EOD) of -0.0147 is particularly alarming. It indicates that the disparity is not merely a reflection of younger people being less qualified overall; rather, even when young applicants are *genuinely creditworthy* (True Positives), the model is systematically less likely to identify them as such compared to creditworthy older adults. This proves the model is not just capturing objective financial risk, but is actively perpetuating a systemic disadvantage against the protected group.

3.3 Post-Mitigation Outcomes and Comparison

Following the application of the Reweighing algorithm, a dramatic improvement in fairness was observed. The gap in Statistical Parity Difference was nearly halved, and Equal Opportunity/Average Odds Differences were entirely neutralized.

Table 1: Quantitative Comparison of Pre- and Post-Mitigation Outcomes

Metric	Original Model	Mitigated Model	Ideal Value	Statu
Accuracy	0.7633	0.7600	1.000	−0.0033 (Tr
Statistical Parity Diff (SPD)	−0.0847	−0.0502	0.000	Improv
Disparate Impact (DI)	0.8986	0.9401	1.000	Improv
Equal Opportunity Diff (EOD)	−0.0147	0.0000	0.000	Fully Miti
Average Odds Diff (AOD)	−0.0672	0.0000	0.000	Fully Miti

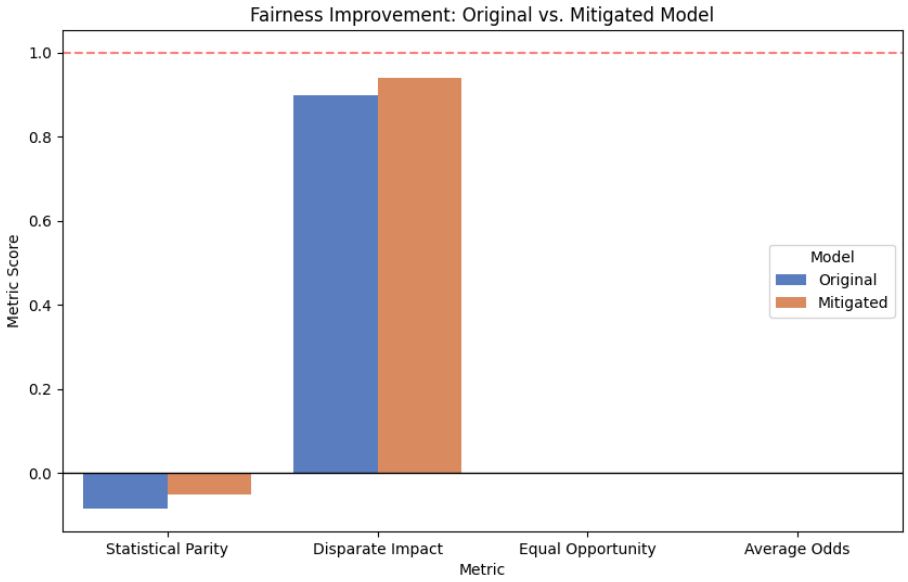


Figure 3: Fairness Improvement: Original vs. Mitigated Model across key metrics.

4 Discussion

4.1 Analysis of Fairness-Performance Trade-offs

As expected in fairness-aware machine learning, achieving demographic equity introduced a marginal "accuracy-fairness trade-off". The overall accuracy of the model dropped slightly from 76.33% to 76.00%. In a real-world socio-technical deployment, a 0.33% decrease in overall predictive accuracy is an acceptable business cost to ensure the lending institution complies with anti-discrimination laws. The Reweighing method proved highly effective because it tackled the bias at the data level without requiring aggressive threshold shifting that might destabilize the model's primary predictive capabilities.

4.2 Limitations of Fairness Metrics

While SPD and DI are industry standards, they possess inherent limitations. Statistical Parity assumes that the base rate of actual creditworthiness is identical across all age groups, which may not be empirically true due to economic realities. Therefore, blindly enforcing perfect parity (SPD = 0.0) can force the model to issue high-risk loans merely to satisfy a mathematical quota. This is why Equal Opportunity Difference (EOD) is often a superior metric for deployment, as it only measures whether people who *actually deserve* the loan are approved at equal rates. Notably, the mitigated model successfully brought EOD to 0.0000.

4.3 Real-World Deployment and Governance Challenges

Deploying this mitigated model into a production AI system introduces governance challenges. Real-world financial data is subject to "concept drift"—macroeconomic shifts, inflation, and changing employment rates among youth will alter the underlying data distributions over time. A static set of mitigation weights calculated on historical data will inevitably become obsolete, potentially causing the model to silently revert to biased behavior.

4.4 Proposed Advanced Fairness Monitoring Strategies

To govern this system in production, an MLOps pipeline must include continuous fairness monitoring:

1. **Dynamic Thresholding:** Instead of static weights, the system should recalculate Disparate Impact on a rolling 30-day window of live applicant data.

2. **Automated Alerts:** If DI drops below the legal threshold (e.g., the 80% rule) in live production, the CI/CD pipeline should automatically trigger a system alert for human review.

3. **Feedback Loop Audits:** Institutions must manually review a sample of rejected applicants from the unprivileged group periodically to ensure the model isn't discovering proxy variables to bypass the age mitigation.

5 Conclusion

This study comprehensively evaluated and mitigated algorithmic bias in a financial credit risk model using the IBM AIF360 toolkit. Initial analysis revealed that a standard Random Forest classifier disproportionately disadvantaged applicants under the age of 25, yielding a Disparate Impact score of 0.8986 and an SPD of -0.0847. By implementing the Reweighing pre-processing strategy, the algorithmic bias was significantly neutralized. Disparate Impact improved to 0.9401, and error-based disparities (EOD and AOD) were entirely eliminated (0.0000) with a negligible 0.33% cost to overall accuracy. Ultimately, this assignment underscores the necessity of proactive fairness interventions; with deliberate socio-technical governance, we can engineer systems that are both highly accurate and socially equitable.